Exercise 5B: Regularisation helps your models converge

In some fields of machine learning, we often use sparse features, i.e., features that only take non-zero values on a small subset of the training set. For instance, in simple natural language processing models, it is common to use *bag-of-words features* for classification. Bag-of-words features are indicator features representing the presence or absence of a specific word from a document:

$$f(w) = \begin{cases} 1 & \text{if the word } w \text{ occurs in the document} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Owing to the distributional characteristics of natural language, it frequently happens that a particular word (say, a certain name or very specific term) only occurs in a single document even in large corpus.

Let us assume that we wish to compute the maximum likelihood estimate of the parameters of a model with the following loss function (negative log-likelihood):

$$L(\mathbf{w}; \mathscr{D}) = \sum_{i=1}^{N} \exp(-y_i(\mathbf{w} \cdot \mathbf{x}_i)) \tag{2}$$

Here, $\mathbf{w}$ is the vector of parameters we are estimating, and $\mathscr{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$ is a dataset of N training examples, each of which consists of a *feature vector* $\mathbf{x}_i$ and a *target label* y_i. This is a *supervised learning* setup, where the features $\mathbf{x}_i$ represent a description of the items in the training set (such as document described by bag-of-words features), and the labels $y_i \in \{-1, 1\}$ represent the intended binary classification of each example into one of two classes represented by the values -1 ("negative") and 1 ("positive"). We would call the model in Eq. 2 a "linear classifier with exponential loss".

For simplicity and to make it possible to do the calculations by hand, we study this example in a very small dataset of only 4 examples.

a) First, assume that our dataset only has two features, so $\mathbf{x}_i, \mathbf{w}_i \in \mathbb{R}^2$. Specifically, this is the training set:

$\mathbf{x}_i$	y_i
$(0,0)$	-1
$(1,0)$	-1
$(0,1)$	-1
$(1,1)$	1

- ▶ Compute the first derivative of the log-likelihood (Eq. 2) with respect to each of the two parameters, i.e., $\partial L/\partial w_1$ and $\partial L/\partial w_2$.
- ▶ Under what conditions can the log-likelihood reach an optimum?

b) Now let us expand the training set with a third, sparse feature that is non-zero only in one of the training examples. The training set now looks like this:

$\mathbf{x}_i$	y_i
$(0,0,1)$	-1
$(1,0,0)$	-1
$(0,1,0)$	-1
$(1,1,0)$	1

- Compute again the derivatives of the log-likelihood with respect to each of the parameters.
- What is the problem if we now try to optimise the log-likelihood by setting the derivatives to zero? (Hint: Look at $\partial L/\partial w_3$)

c) Now add an ℓ_2 regularisation term to Eq. 2 and recompute the derivatives. Remember that the ℓ_2 regularisation term is the squared 2-norm

$$||\mathbf{w}||_2^2 = \sum_{i=1}^{d} w_d^2, \tag{3}$$

where d is the dimensionality of the parameter vector.

- Under what conditions can the log-likelihood now reach an optimum?